TP4: OpenStack Installation Guide (Ubuntu 24.04 LTS + DevStack)

Purpose: Install an OpenStack environment for Lab 1 (service exploration) and Lab 2 (virtual networking and VM deployment).

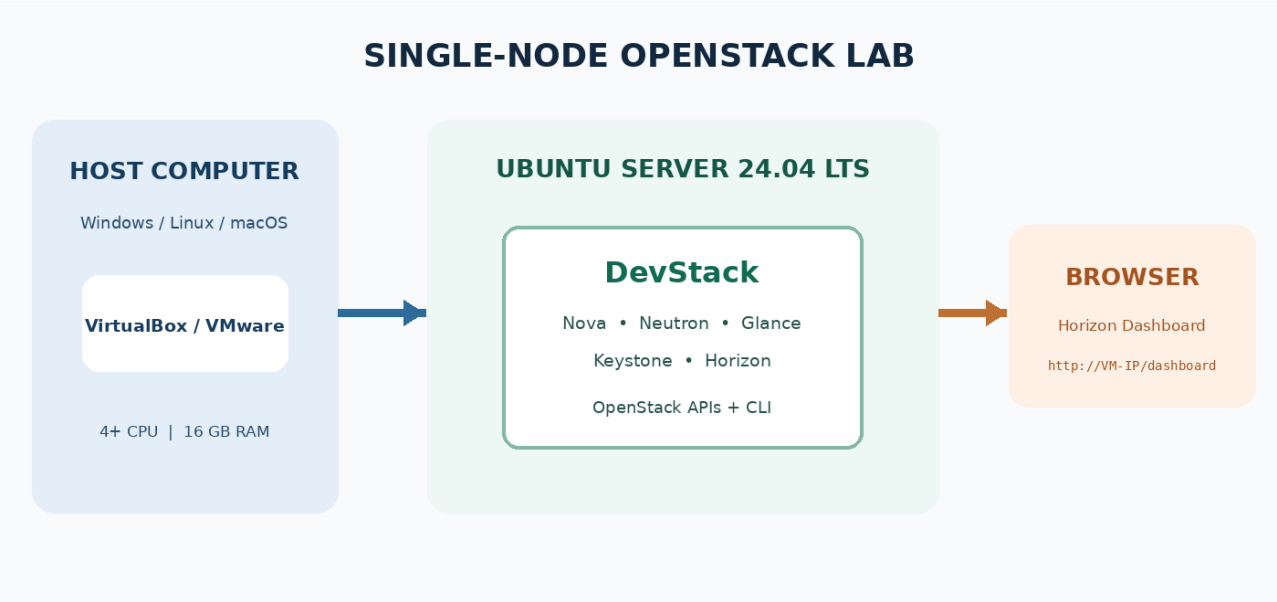


Figure 1. Overview of the proposed laboratory environment (original diagram).

1. Before you begin

DevStack is intended for development and educational testing, not a production cloud. It modifies operating-system and networking configuration. Use a dedicated Ubuntu installation or a disposable virtual machine.

Operating system	**Ubuntu Server 24.04 LTS, clean installation**
Suggested allocation	4+ vCPUs; 16 GB RAM; 80 GB disk
Network	Internet access; bridged networking can simplify browser access
Privileges	One Ubuntu account with sudo access
Virtualization	Enable VT-x/AMD-V, and nested virtualization when supported

The resource allocation above is a classroom recommendation, not an official minimum. A VM without nested virtualization may run guest instances slowly using software emulation.

2. Create the Ubuntu virtual machine

In VirtualBox or VMware, create a new Linux/Ubuntu 64-bit VM, attach the Ubuntu Server 24.04 LTS installation ISO, allocate resources, and complete the installer. Choose a username and password you can use for sudo. Enable an internet-connected network adapter.

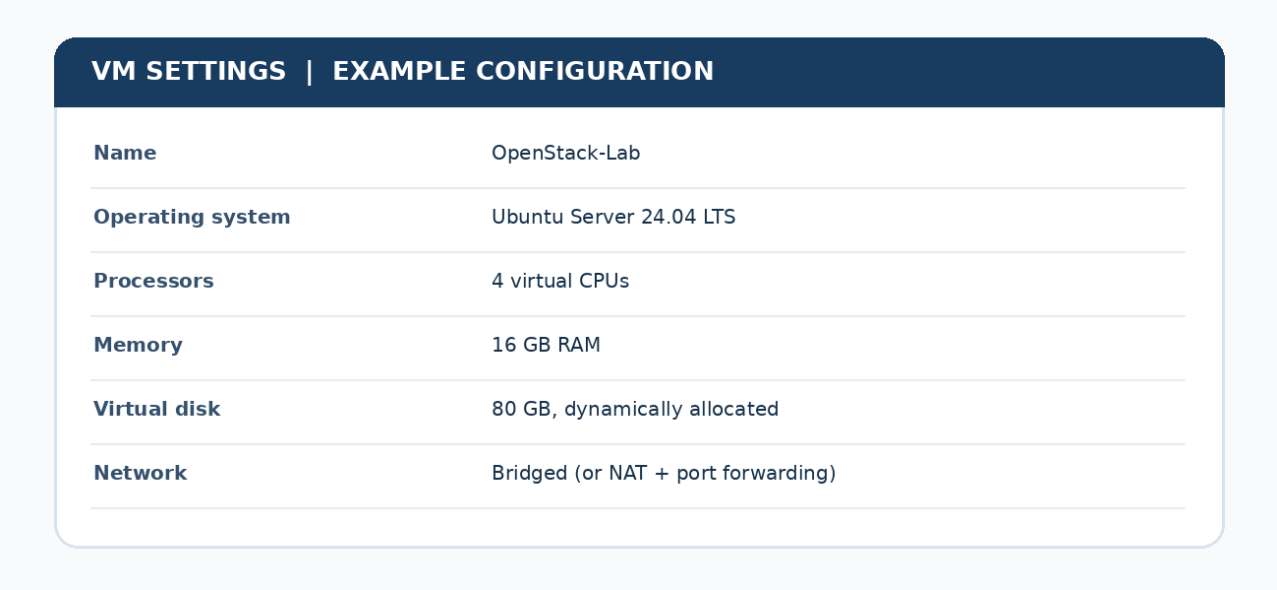


Figure 2. Recommended example virtual-machine settings (illustration).

After Ubuntu starts, log in and verify the IP address and internet connectivity:

```
hostname -I
ping -c 3 archive.ubuntu.com
```

3. Installation sequence

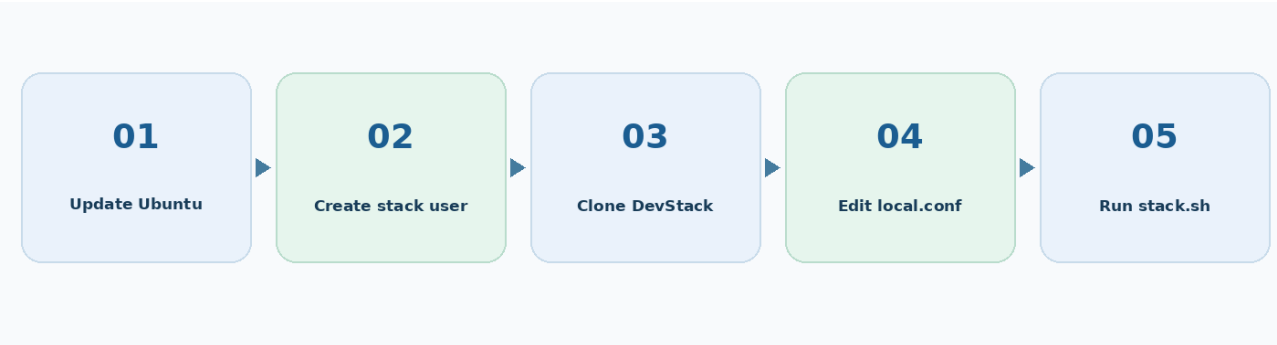


Figure 3. Five main stages of a DevStack deployment (original diagram).

Step 1. Update Ubuntu

```
sudo apt update
sudo apt upgrade -y
```

```
sudo apt install -y git curl wget
```

Step 2. Create a non-root stack user
DevStack must be launched by an ordinary user with passwordless sudo permission. These commands create a dedicated account for this disposable lab machine.

```
sudo useradd -s /bin/bash -d /opt/stack -m stack
sudo chmod +x /opt/stack
echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/stack
sudo -u stack -i
```

After the last command, the terminal should be operating as user stack. Do not use sudo to execute stack.sh.

Step 3. Download DevStack

```
git clone https://opendev.org/openstack/devstack
cd devstack
```

The files are now located in /opt/stack/devstack when using the stack account configured above.

Step 4. Create the configuration file
Create local.conf inside the DevStack directory:

```
nano local.conf
```

Enter the following minimal configuration. Change the example password to a unique alphanumeric value:

```
[[local|localrc]]
ADMIN_PASSWORD=LabCloud2026
DATABASE_PASSWORD=$ADMIN_PASSWORD
RABBIT_PASSWORD=$ADMIN_PASSWORD
SERVICE_PASSWORD=$ADMIN_PASSWORD
```

Save using Ctrl+O, Enter, then Ctrl+X. The [[local|localrc]] heading is part of the file, not a shell command. Avoid punctuation in these example passwords because DevStack warns that some services can fail with special characters.

Step 5. Start the deployment

```
./stack.sh
```

The script installs dependencies and configures multiple OpenStack components. A successful run prints OpenStack/Horizon access information. Installation time depends on downloads, disk, and processor performance. If an error occurs, read the terminal output rather than assuming the installation succeeded.

4. Access the Horizon dashboard

Get the IP address of the Ubuntu machine:

```
hostname -I
```

Open the Horizon URL reported by stack.sh in a browser. In many DevStack setups, the format is http://<VM-IP>/dashboard. Sign in with username admin and the ADMIN_PASSWORD configured in local.conf, using the Default domain if requested.

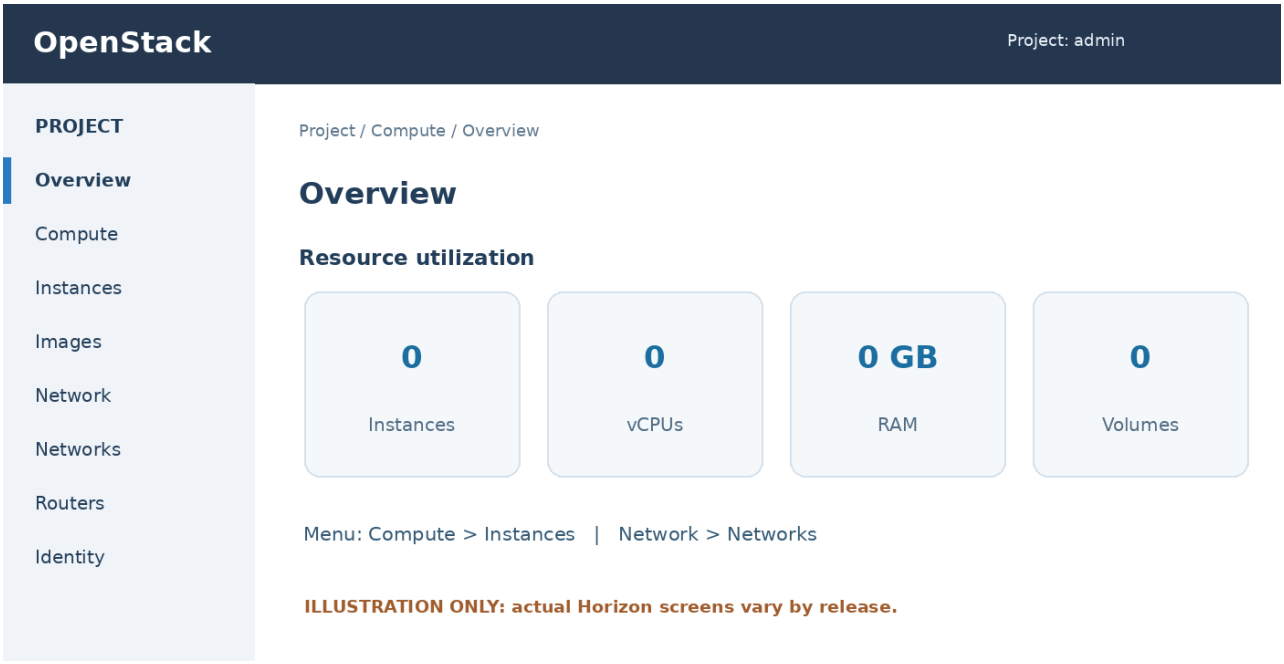


Figure 4. Simplified Horizon dashboard layout. This is an illustrative mock-up, not a captured live screen.

If the page is unreachable, verify that the VM networking mode permits browser access. For NAT, port forwarding may be necessary; alternatively use bridged networking on a trusted local network.

5. Verify the installation from the CLI

In the Ubuntu terminal, authenticate as the administrator:

```
cd /opt/stack/devstack
source openrc admin admin
```

Run the following commands to inspect the deployed environment:

```
openstack token issue
openstack service list
openstack image list
openstack flavor list
openstack network list
openstack server list
```

Expected outcome: token issue returns token metadata; service list shows registered services; the list commands return tables, although some lists may be empty. These outputs, along with a working Horizon login, indicate that core installation checks succeeded.

6. OpenStack components used in the labs

Keystone	**Authentication and identity**
Nova	Compute instance management
Neutron	Networking, subnets, routing, security groups
Glance	Images for virtual machines
Cinder	Block volumes
Horizon	Web-based graphical interface
Placement	Compute resource inventory and allocation

7. Common problems and solutions

Permission error	**Run the installer as stack, not root; verify sudo privileges.**
No internet / download errors	Check DNS, network adapter, proxy, and outbound internet access.
Not enough resources	Increase allocated RAM or disk space and ensure sufficient host resources.
No Horizon access	Use the actual URL printed by stack.sh and verify VM connectivity.
Slow VM instances	Check nested hardware virtualization; guest instances may be emulated.
OpenStack CLI authentication fails	Reload source openrc admin admin and check the installation output.

8. Lab checklist

☐ Ubuntu installed and has internet access

☐ Dedicated stack user created

☐ DevStack repository cloned

☐ local.conf created with passwords

☐ stack.sh completed successfully

☐ Horizon login successful

☐ OpenStack CLI service list works

☐ Ready for Lab 1 and Lab 2

9. Official references

DevStack Quick Start: https://docs.openstack.org/devstack/latest/

Single-VM DevStack guide: https://docs.openstack.org/devstack/latest/guides/single-vm.html

OpenStack CLI documentation: https://docs.openstack.org/python-openstackclient/latest/

Ubuntu Server download: https://ubuntu.com/download/server

Note: The figures in this document are instructional illustrations. The precise interface, IP addresses, and displayed results depend on your installation and DevStack release.